

Introduction

Electromagnetic induction is the process of generating an electric current in a conductor by changing the magnetic field around it. This phenomenon was discovered by Michael Faraday and is the basic principle behind transformers, generators, and many electrical devices.

Faraday's Laws of Electromagnetic Induction

1. **First Law:**

Whenever there is a change in magnetic flux linked with a circuit, an electromotive force (EMF) is induced in the circuit.

2. **Second Law:**

The magnitude of induced EMF is directly proportional to the rate of change of magnetic flux.

Formula:

$$\text{EMF} = -\frac{d\Phi}{dt}$$

(The negative sign indicates Lenz's Law.)

Lenz's Law

Lenz's Law states that the direction of induced current always opposes the cause that produces it. This law ensures the conservation of energy.

Applications of Electromagnetic Induction

- Electric generators
 - Transformers
 - Induction cooktops
 - Electric motors
 - Wireless charging systems
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Example Problem

A coil has a magnetic flux changing from 0.02 Wb to 0.01 Wb in 0.5 seconds. Calculate the induced EMF.

Solution:

$$\text{EMF} = \frac{\Delta \Phi}{\Delta t} = \frac{0.01 - 0.02}{0.5} = -0.02 \text{ V}$$
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Magnitude of EMF = **0.02 V**

Conclusion

Electromagnetic induction is a fundamental concept in physics that plays a vital role in modern technology. Understanding this concept helps in explaining how electrical energy is generated and used efficiently.